HOPE: Hierarchical Fusion for Optimized and Personality-Aware Estimation of Depression

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Abstract

Depression detection remains challenged by generalized modeling approaches that fail to account for individual heterogeneity. To address this, the Multimodal Personality-aware Depression Detection (MPDD) Challenge introduced personalized features into the modeling process, aiming to better capture individual variability. However, the baseline models still exhibit two critical limitations: the neglect of textual semantics embedded in audio, and inconsistent predictions for the same subject across tasks and samples. Motivated by these limitations, we introduce **HOPE** (**H**ierarchical fusion for **O**ptimized and **P**ersonality-aware **E**stimation of Depression), a unified framework for consistent, subject-level depression estimation. HOPE first employs a Latent Semantic Projection (LSP) module to reconstruct textual semantics from audio features when transcripts are unavailable. It then introduces a consistency-aware integration mechanism that hierarchically fuses multi-branch predictions to resolve inter-task and inter-sample contradictions. HOPE achieved first place in the MPDD Challenge Young Track, demonstrating strong cross-modal learning capabilities and consistent, subject-level depression prediction.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**.

Keywords

Depression Detection; Multimodal Learning; Personality-aware Modeling; Hierarchical Fusion

ACM Reference Format:
Hanlei Shi, Yu Liu, Haoxun Li, Yuxuan Ding, Jiaxi Hu, Leyuan Qu, and Taihao Li. 2025. HOPE: Hierarchical Fusion for Optimized and Personality-Aware Estimation of Depression. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3746027.3762063

1 Introduction

Depression is a mental health disorder and has emerged as a major public health concern worldwide [15, 22]. This disorder not only severely impairs an individual's quality of life and social functioning but also poses long-term risks to their physical health, family relationships, and professional development [16, 19]. Traditionally, depression assessment methods have mainly relied on psychometric scales [1, 8] and shallow learning algorithms applied to structured data [4, 21]. In recent years, the rapid progress of deep learning has opened new directions for depression detection, especially through multimodal approaches that integrate information from speech, facial expressions, and text data [2, 6, 24].

However, most existing approaches still adopt a *generalized modeling* paradigm which often fail to account for the heterogeneity of depression across individuals [13, 18]. For instance, even at the same level of depression severity, people with different personality

*These authors contributed equally to this work.
Hanlei Shi: Designed and implemented the modified multi-branch architecture, with specialized integration of the LSP module. **Yu Liu**: Developed the consistency-aware subject-level fusion strategy; led the writing and completed final code integration and open-source release preparation. **Haoxun Li**: Proposed and implemented the LSP module, including its architecture design, feature alignment strategy, and supervised training pipeline. **Yuxuan Ding**: Procured and preprocessed MER datasets; conducted component-wise analysis attributing gains to innovations. Author order was randomized to reflect equal contribution.
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MM '25, Dublin, Ireland
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ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3762063

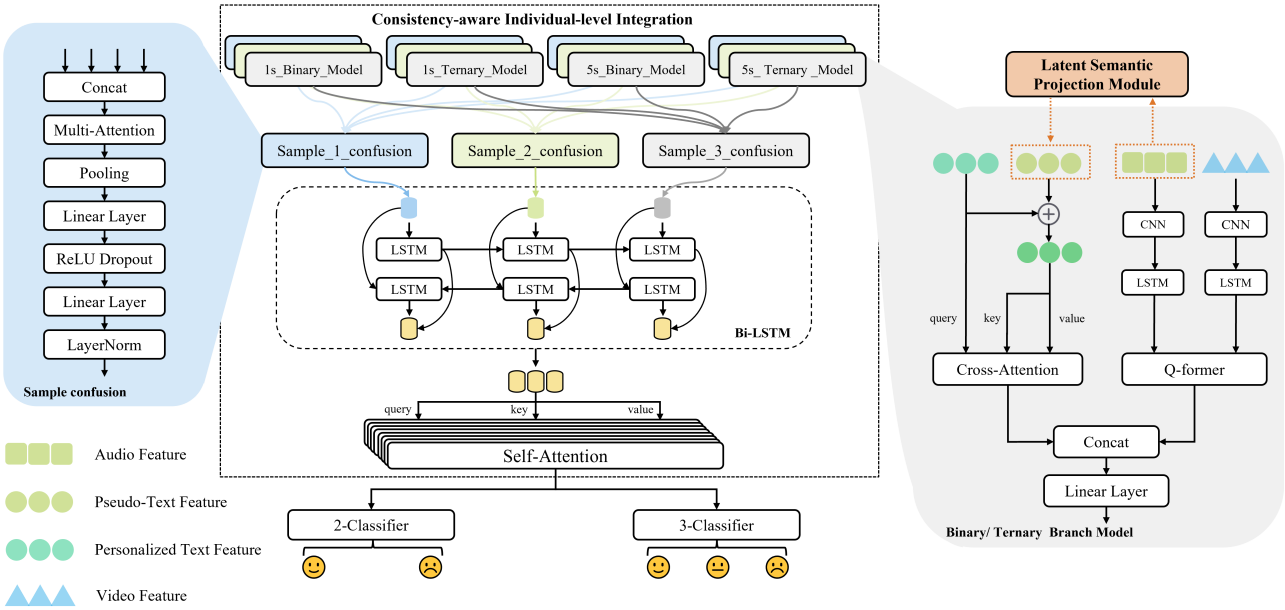

Figure 1: The overview of HOPE. We first employ a LSP module to extract corresponding textual semantic features from the audio. In the first stage, four modal features (video, audio, personalized text, and pseudo-text feature) are fed into the branch model for preliminary fusion to enhance the capability of individual branches. During the second stage, sample confusion is conducted based on the branch model outputs from the first-stage, followed by secondary feature fusion. The system ultimately generates both binary and ternary classification results through a dual-classifier architecture.

traits may display distinct patterns in their speech prosody, facial cues, or linguistic style [23]. Due to individual variability, current depression detection models exhibit limited performance [3, 7, 14].

To highlight these limitations, the ACM Multimedia 2025 MPDD Challenge [5] introduced a benchmark task based on pre-extracted multimodal features. While the baseline MPDD models have validated the critical role of personality traits in depression modeling and demonstrated that incorporating personality information significantly improves prediction accuracy, two key limitations persist. First, the baseline models ignore the textual semantic content conveyed by the audio (i.e., the actual verbal expressions of the subjects), which often contains crucial clinical cues for identifying depressive tendencies. Second, in the MPDD dataset, each subject is associated with three distinct samples, each evaluated under two window lengths (1s and 5s) and two classification tasks (binary and ternary). In the baseline setting, these are treated as independent instances, disregarding subject identity. As a result, existing methods often produce inconsistent and contradictory clinical predictions across samples, window lengths, and tasks for the same individual.

To address these limitations, we propose **HOPE—Hierarchical fusion for Optimized and Personality-aware Estimation of depression**, a unified framework tailored for consistency-aware, subject-level depression prediction. The innovations of this study are as follows: first, inspired by the observation that textual information in audio-visual dialogues can provide richer text semantic features [17, 19], we construct a novel LSP module trained on extensive external dataset. In scenarios where raw dialogue transcripts are unavailable and access to original audio-visual data is restricted, this module reconstructs textual semantic representations from audio features, effectively supplementing critical text-based semantic information. Second, to address the subject-level prediction inconsistency observed in the MPDD baseline, we design a consistency-aware individual-level integration mechanism. By hierarchically fusing outputs from multiple task-specific and sample-specific branches, the model learns to resolve logical contradictions and amplify the most reliable signals, ensuring coherent and clinically plausible subject-level predictions. The main contributions of this study can be summarized as follows:

- We propose the LSP module, which extracts transcript-level semantic information from audio features when the raw dialogue text data and audio-visual data are inaccessible, thereby supplementing critical text-based semantic information carried by the audio.
- We propose a consistency-aware individual-level integration mechanism that hierarchically fuses multi-branch predictions to resolve inter-task and inter-sample contradictions in the baseline of MPDD, thereby enhancing the subject-level reliability of depression assessment.
- Our proposed HOPE framework secured first place in the MPDD-Young competition track, demonstrating superior cross-modal learning capabilities under strict privacy constraints (only pre-extracted features were permitted) and limited training samples.

2 Method

2.1 Method Overview

To address two limitations of the baseline which neglect audio's textual semantics (containing clinically vital verbal expressions) and which yield inconsistent judgments by processing same-subject samples independently, we propose HOPE. As shown in Fig. 1, its framework contains LSP Module and consistency-aware individual-level integration mechanism , specifically addressing these issues.

Inspired by evidence that textual modalities provide richer clinical cues than audio-visual data for depression detection [17, 19], we trained the LSP Module on the auxiliary MER2025 dataset [10] to establish end-to-end mapping from audio to textual features—enabling direct semantic extraction without speech transcripts [9]. The resulting "Pseudo-Text Features" are incorporated as a new modality into our framework, with technical implementations detailed in Section 2.2.

While preserving the four-branch subtask structure defined by the MPDD challenge, HOPE builds on the baseline design where separate feature extractors are used for each window length (1s, 5s) and task type (binary, ternary), forming four distinct branches. Each subject contributes 12 inputs in total—three samples under two window lengths and two task settings. However, the baseline treats these inputs independently, often yielding conflicting predictions across tasks and temporal windows for the same subject. To resolve this, we introduce a two-stage hierarchical fusion mechanism that promotes subject-level consistency. Sample outputs are first aggregated into task-level embeddings via learned attention, then integrated through a temporal reasoning module that reconciles discrepancies across branches. This allows the model to selectively amplify reliable predictions, producing coherent and clinically meaningful outcomes at the subject level. Further technical details are elaborated in Section 2.3, and Section 2.4.

2.2 Latent Semantic Projection Module

In addition to leveraging static personality traits, HOPE introduces a dynamic semantic representation derived from the subject's spoken audio. This representation is designed to encode the high-level linguistic meaning—complementing what it was said (content) and by whom (personality). The goal is to project each audio segment into a semantic space aligned with sentence-level language models, enabling deeper understanding of depression-relevant content.

Due to privacy protection constraints in the MPDD, only pre-extracted Wav2Vec [20] features are available instead of raw audio data. To extract richer semantic representations from these limited audio features that better capture textual content, as shown in Fig 2, we design a pooled mapping approach that transforms the available 512-dimensional audio features into a text-aligned semantic space of 1024 dimensions.

Our approach, formalized in Algorithm 1, adopts a streamlined training strategy that treats the entire utterance as a unit during training while applying the learned mapping segment-wise during inference. Specifically, during the training phase, we average-pool the full Wav2Vec feature sequence from each utterance to produce a global 512-dimensional representation. This pooled feature is then processed through a three-layer feedforward network f_θ with intermediate dimensions of 1024, incorporating ReLU activations, dropout ($p = 0.2$), and layer normalization. The network is trained using cosine similarity loss to minimize the semantic distance between predicted features and ground-truth RoBERTa [12] sentence embeddings.

At inference time, we apply the trained mapping network f_θ to each 5-second audio segment of the MPDD data independently. As detailed in Algorithm 1, given segment-level input features of shape $(T, 512)$ where each segment corresponds to a 5-second audio chunk, our model outputs a sequence of semantic vectors

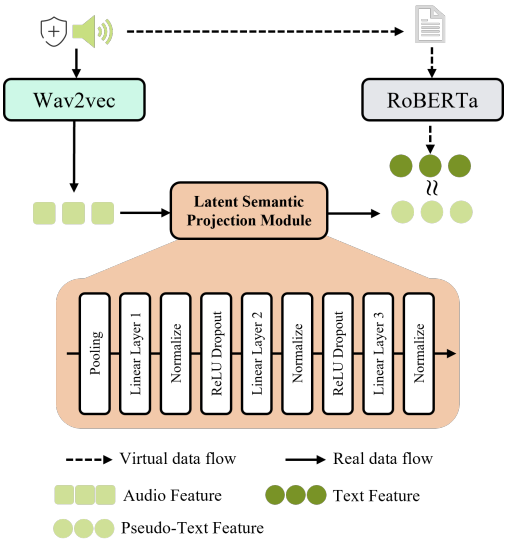

Figure 2: Illustration of the feature mapping process. The LSP module converts audio features into corresponding textual semantic features. The Virtual Data Flow represents the processing pipeline when raw audio data is available, while the Real Data Flow denotes the scenario where only audio features are used.

Algorithm 1 Pseudo Code of LSP Module

Input: Training pairs $\{(a_i, t_i)\}$, where $a_i \in \mathbb{R}^{512}$, $t_i \in \mathbb{R}^{1024}$; MPDD audio feature $A_{comp} \in \mathbb{R}^{T\times512}$ (5-second).
Output: Semantic sequence $T_{semantic} \in \mathbb{R}^{T\times1024}$
Training Phase:
repeat
  Sample (a, t) from training set
  $t_{pred} \leftarrow f_\theta(a)$
  $L \leftarrow 1 - \cos(t_{pred}, t)$
  $\theta \leftarrow \theta - \alpha\nabla_\theta L$
until convergence
Inference Phase (MPDD Data):
$T_{semantic} \leftarrow [\,]$
for $j = 1$ **to** T **do**
  $t_j \leftarrow f_\theta(A_{comp}[j])$
  $T_{semantic}$.append(t_j)
end for
return $T_{semantic} \in \mathbb{R}^{T\times1024}$

with shape $(T, 1024)$. Each output vector captures the latent linguistic semantics of the corresponding audio segment, creating a temporally-aligned semantic representation that preserves both the temporal structure of the original audio and the semantic richness of the corresponding textual content.

These generated pseudo-text features provide the Keys and Values in a cross-attention [11] mechanism, while the personalized texts serve as Queries. This interaction allows the model to align *what* the subject expresses with *who* they are, encouraging the extraction of highly personalized, depression-relevant cues. This attention mechanism operates exclusively at the segment level, producing personality-conditioned representations that serve as fixed inputs to the subsequent hierarchical fusion module.

Importantly, the resulting pseudo-text features are treated as an additional text-aligned modality, orthogonal to raw audio features, and seamlessly integrated into the multimodal fusion process. Empirical results in Section 3 confirm that this design leads to improved discrimination in both binary and ternary settings.

2.3 Consistency-aware Individual-level Feature Integration

Building upon the MPDD baseline, each of the four branches in our HOPE framework corresponds to a distinct temporal resolution and classification task. To enhance feature quality under the strict input constraints of the challenge, we introduce three lightweight and effective modifications: (i) convolutional layers to capture local audio–visual patterns, (ii) a Q-Former module for cross-modal alignment between audio and visual features, and (iii) a personality-aware attention mechanism that injects subject-specific priors into sample-level representations.

These enriched embeddings are used to derive a stable and consistent representation for each subject through a two-stage fusion strategy, which integrates features across both subtasks and temporal samples in a $12 \rightarrow 3 \rightarrow 1$ pattern. This layered integration enables the model to consolidate heterogeneous cues into a coherent individual-level semantic representation. Unlike conventional attention-based fusion that assumes feature compatibility, our approach **shifts the paradigm from feature aggregation to conflict arbitration**, explicitly addressing cross-task prediction conflicts through personality-aware resolution.

At the stage one, we aggregate information from four distinct modality-specific branches—audio, video, personalized text, and pseudo-text—each of which is processed by a corresponding subnetwork within the Binary/Ternary Branch Model (see Fig. 1). These sub-networks transform their respective inputs into intermediate representations, denoted as $\mathbf{f}_{branch}^{(i)}$. To enable effective fusion, each branch-specific feature vector is linearly transformed into a shared latent space:

$$\mathbf{f}_{aligned}^{(i)} = \text{Linear}_i(\mathbf{f}_{branch}^{(i)}) \in \mathbb{R}^{512}, \quad i \in \{1, 2, 3, 4\} \tag{1}$$

The aligned vectors are stacked into a tensor of shape 4×512 and passed through a multi-head attention (MHA) module to model inter-branch dependencies:

$$\mathbf{F}_{attn} = \text{MHA}(\mathbf{F}_{stacked}, \mathbf{F}_{stacked}, \mathbf{F}_{stacked}) \tag{2}$$

where $\mathbf{F}_{stacked} = \text{stack}([\mathbf{f}_{aligned}^{(1)}, \ldots, \mathbf{f}_{aligned}^{(4)}])$. The output from the attention layer is averaged across branches and enhanced using a two-layer MLP with residual connections and layer normalization:

$$\mathbf{f}_{fused} = \text{LayerNorm}(\text{MLP}(\text{mean}(\mathbf{F}_{attn}))) \in \mathbb{R}^{512} \tag{3}$$

This fused vector captures a consistent subtask-level embedding at a given time step, combining signals across temporal scales and classification granularities.

The stage two of fusion integrates temporal information across the three input windows. As illustrated in Fig. 1, we obtain three fused vectors $\mathbf{f}_{fused}^{(1)}, \mathbf{f}_{fused}^{(2)}, \mathbf{f}_{fused}^{(3)}$ corresponding to different samples for each subject. These are fed into a BiLSTM to encode temporal dependencies:

$$\mathbf{H}_{temporal} = \text{BiLSTM}([\mathbf{f}_{fused}^{(1)}, \mathbf{f}_{fused}^{(2)}, \mathbf{f}_{fused}^{(3)}]) \in \mathbb{R}^{3\times 512} \tag{4}$$

We apply another MHA layer over the temporal outputs to capture varying importance across samples:

$$\mathbf{H}_{attn}, \mathbf{W}_{attn} = \text{MHA}(\mathbf{H}_{temporal}, \mathbf{H}_{temporal}, \mathbf{H}_{temporal}) \tag{5}$$

Attention weights are averaged across heads and normalized with a softmax function to compute the temporal importance distribution:

$$\mathbf{w}_{temporal} = \text{SoftMax}(\text{mean}(\mathbf{W}_{attn}, \text{dim} = 1)) \tag{6}$$

Finally, a weighted sum of the attended temporal features is computed, followed by an MLP to produce the final subject-level representation:

$$\mathbf{f}_{final} = \text{MLP}\left(\sum_{t=1}^{3} \mathbf{w}_{temporal}^{(t)} \cdot \mathbf{H}_{attn}^{(t)}\right) \in \mathbb{R}^{512} \tag{7}$$

The resulting vector $\mathbf{f}_{final}$ encodes semantically rich, temporally aware, and personality-aligned information. It serves as the input for downstream multi-task classification, enabling coherent binary and ternary predictions per subject.

2.4 Multi-task Classification

To support nuanced depression assessment, HOPE adopts a multi-task learning strategy that simultaneously predicts both binary (healthy vs. depressed) and ternary (healthy vs. mild vs. severe) labels. This formulation encourages the model to learn shared representations that capture core depressive signals while allowing task-specific classifiers to focus on different severity granularity.

Both classification heads operate on the same final feature vector $\mathbf{f}_{final} \in \mathbb{R}^{512}$, which encodes personality-aware, temporally integrated, and multimodally aligned information. We adopt the same fully connected classification architecture as used in the MPDD baseline for both heads. The binary classifier C_{binary} outputs logits for two classes, while the ternary classifier $C_{ternary}$ produces logits over three severity levels. Given the shared feature input $\mathbf{h}_{shared} = \mathbf{f}_{final}$, the predicted distributions are:

$$P(y_{binary}) = \text{SoftMax}(C_{binary}(\mathbf{h}_{shared})), \tag{8}$$

$$P(y_{ternary}) = \text{SoftMax}(C_{ternary}(\mathbf{h}_{shared})) \tag{9}$$

The overall training objective is a weighted combination of binary and ternary cross-entropy losses:

$$\mathcal{L}_{total} = \alpha_b \cdot \mathcal{L}_{binary} + (1 - \alpha_b) \cdot \mathcal{L}_{ternary} \tag{10}$$

where α_b is the binary-task weighting factor. We empirically set $\alpha_b = 0.5$, which yielded the best overall performance across both tasks on the validation set. This balanced weighting ensures that neither task dominates the learning process, promoting effective representation sharing and mutual reinforcement. Both losses are computed using standard categorical cross-entropy:

$$\mathcal{L}_{binary} = -\sum_{i=1}^{N} y_i^{binary} \log P(y_i^{binary}), \tag{11}$$

$$\mathcal{L}_{ternary} = -\sum_{i=1}^{N} y_i^{ternary} \log P(y_i^{ternary}) \tag{12}$$

3 Experiment

3.1 Dataset and metrics

LSP module dataset and metrics. To train and evaluate the proposed LSP module, we utilize the MER2025 dataset—a large-scale multimodal dataset developed for emotion recognition in real-world Chinese media content [10]. The full dataset contains more than 180,000 utterances, each annotated with corresponding emotional labels and aligned across audio, text, and video modalities. For our purposes, we focus exclusively on paired audio-text samples, selecting 100,000 utterances for training the LSP module. These are split into training, validation, and test sets in a 7:1.5:1.5 ratio. To evaluate the quality of the projected Pseudo-Text Features, we employ Root Mean Square Error (RMSE) and Cosine Similarity (Cosine Sim) as complementary metrics.

Full model dataset and metrics. We evaluate our full model on the MPDD-Young dataset [5], a multimodal dataset designed for depression detection in young adults. This dataset provides synchronized audio, visual, and demographic data, alongside PHQ-9-based depression severity labels and personality attributes based on the BigFive-10 scale. Each sample is further enriched with gender, age, and birth region metadata to support more nuanced analysis. Instead of raw audio or video, the organizers provide pre-extracted multimodal features for each participant. Each individual has three data samples per modality (denoted as _1, _2, _3). The audio, visual, and personalized semantic features used in our experiments follow the configuration provided in the MPDD-Young dataset [5].

Following the baseline evaluation protocol, we adopt four metrics: weighted and unweighted F1 scores (W_{F_1} / U_{F_1}) as well as weighted and unweighted accuracies (W_{Acc} / U_{Acc}). The final score is computed as a weighted average of the F1 scores and accuracies across the four tasks.

In our evaluation, each task$_i$ in the above equations corresponds to a specific classification setup supported by the dataset. The MPDD-Young dataset defines two tasks: (1) binary classification between *normal* and *depressed*, and (2) ternary classification among *normal*, *mildly depressed*, and *severely depressed*. Performance metrics are averaged across these tasks to compute the final track-level score.

3.2 Experimental Setup

LSP module experimental setup. We compare two strategies for generating pseudo-text features from audio inputs, distinguished primarily by the structure of their outputs. The first approach produces a single 1024-dimensional vector for each utterance, while the second—our adopted strategy—yields a sequence of vectors with shape $(T, 1024)$, where T denotes the number of 5-second segments in the utterance.

Table 1: LSP Result Of Different Mapping Model

Model	Text encoder	Cosine Sim ↑	RMSE ↓
Map1	Bert	87.85	51.50
	RoBERTa	95.63	33.82
Map2	Bert	87.84	46.62
	RoBERTa	95.63	25.46

In the first method called Map1, each utterance is divided into 5-second chunks, from which 512-dimensional Wav2Vec2 features are extracted and stacked into a sequence of shape $(T, 512)$. This sequence is then passed through a Transformer encoder prepended with a $[cls]$ token, and the output corresponding to $[cls]$ is trained to regress the RoBERTa embedding of the full transcription. In contrast, our approach processes each 5-second segment independently using a trained mapping network, which we called Map2, producing a sequence of segment-level semantic embeddings with shape $(T, 1024)$. This design retains the temporal dynamics inherent in the pre-extracted audio features and enables the capture of fine-grained semantic cues over time. Such a representation is inherently compatible with our hierarchical fusion mechanism, which operates across time and modality.

Full model experimental setup. Our full model's experiments adopt a two-stage training strategy: (1) The first stage focuses on training each branch model to specialize in depression detection for its specific input type—defined by a particular combination of sample, window length, and classification task—thereby enhancing task-specific performance; (2) The second stage builds upon the trained branch models from the first stage, employing a hierarchical fusion mechanism that integrates task-specific features across branches and temporal samples.

The first stage adopts a 10-fold cross-validation strategy following the baseline configuration, where each of the four task-specific branches is trained separately using parameters consistent with the original setting. The second stage inherits the overall configuration from the first stage—including data loading strategies and random seed settings—and is trained for 100 epochs using the best-performing fold's weights. For all experiments, we use a batch size of 8 and fix the random seed. The dataset is split by subject IDs to prevent identity leakage between training and validation. The network employs ReLU activations, with a dropout rate of 0.3 applied to the classifier layers. Optimization is conducted using Adam (learning rate = 0.001, β_1 = 0.9, weight decay = 1e-4), and a plateau-based scheduler reduces the learning rate when validation accuracy stagnates. All models are implemented in PyTorch 2.3.0 and trained on 8×L20 GPUs.

3.3 Experimental Results

Feature mapping comparison. As shown in Table 1, RoBERTa consistently outperforms BERT across both mapping strategies, indicating better suitability for our semantic projection task. Both

Table 2: Results on the validation set with 1-second and 5-second windowed inputs. The symbols ∗ , † , and ‡ indicate that the corresponding models are built upon the best-performing configuration from the previous group within the table (as highlighted by the light blue row).

Model	1s/5s Binary				1s/5s Ternary			
	U_{F_1}	W_{F_1}	$U_{Acc.}$	$W_{Acc.}$	U_{F_1}	W_{F_1}	$U_{Acc.}$	$W_{Acc.}$
Baseline	49.77/54.64	49.77/54.64	50.00/54.92	50.00/54.92	35.05/45.75	41.60/54.64	41.60/54.65	34.85/46.09
Baseline + GRU	60.10/60.59	60.10/60.59	60.23/60.61	60.23/60.61	38.77/44.81	45.99/53.51	45.83/53.41	38.67/45.04
Baseline + CNN	68.18/67.04	68.18/67.04	68.18/67.05	68.18/67.05	45.48/44.75	53.07/51.02	52.57/51.14	46.34/45.30
∗ + Mix up	74.88/74.99	74.88/74.99	75.00/75.00	75.00/75.00	51.95/58.56	58.74/64.46	57.95/64.77	51.55/57.23
∗ + SMOTE	54.52/59.42	54.52/59.42	54.55/59.47	54.55/59.47	35.05/45.75	41.60/54.64	41.67/54.55	34.85/46.09
∗ + Class-weighted	70.44/71.56	70.44/71.56	70.45/71.59	70.45/71.59	49.27/52.55	55.77/59.38	55.68/59.09	49.46/52.59
∗ + Border-SMOTE	78.34/78.34	78.34/78.34	78.41/78.41	78.41/78.41	59.35/58.21	64.35/64.93	63.64/64.77	59.66/57.23
∗ + Mix up+SMOTE	65.86/63.47	65.86/63.47	65.91/63.64	65.91/63.64	42.00/42.13	47.32/50.14	47.73/50.00	40.32/42.20
∗ + Mix up+Border-SMOTE	75.98/73.86	75.98/73.86	76.14/73.86	76.14/73.86	52.95/55.70	59.96/61.22	59.09/61.36	52.59/54.10
∗ + Mix up+Class-weighted	69.31/68.16	69.31/68.18	69.32/68.18	69.32/68.18	45.96/49.27	55.42/55.77	55.68/55.68	45.99/49.46
† + Concat	77.17/76.06	77.17/76.06	77.27/76.14	77.27/76.14	57.46/63.39	63.90/66.35	63.64/65.91	56.47/62.03
† + Q-former	81.73/82.95	81.73/82.95	81.82/82.95	81.82/82.95	71.68/67.44	75.44/72.51	76.14/72.73	76.14/66.57
‡ + Map1 BERT Concat	80.66/82.90	80.66/82.90	80.68/82.95	80.68/82.95	62.24/72.69	67.44/76.62	67.05/77.27	62.22/77.27
‡ + Map1 RoBERTa + Concat	86.36/86.36	86.36/86.36	86.36/86.36	86.36/86.36	80.91/78.97	81.72/80.57	81.82/80.68	81.82/80.68
‡ + Map2 RoBERTa + Concat	88.63/92.04	88.63/92.04	88.64/92.05	88.64/92.05	82.75/76.00	82.83/81.64	82.95/81.82	82.95/75.51
‡ + Map2 RoBERTa + Q-former	91.64/95.45	91.64/95.45	91.67/95.45	91.67/95.45	85.05/78.77	85.71/85.12	85.98/85.23	85.98/78.35
‡ + Map2 RoBERTa + Cross-Att	96.21/97.73	96.21/97.73	96.21/97.73	96.21/97.73	86.09/80.71	86.93/87.53	87.12/87.50	87.12/80.43

methods achieve comparable cosine similarity scores when using the same text encoder, suggesting that semantic directionality is generally well preserved. However, our method yields a notably lower RMSE under RoBERTa supervision, highlighting its advantage in preserving the numerical fidelity of semantic features.

We innovatively designed two audio-text feature mapping architectures and applied the generated features to the optimal Stage One model for comparative validation. Experimental data in Table 1 clearly demonstrates that the novel feature mapping approach exhibits competitive performance. The final implement of LSP module is recording to "Map2 RoBERTa" settings according to the result shows in Table 1 and Table 2.

Branch model results. (1) To capture local audio-visual patterns, we augmented the baseline's LSTM feature extraction with additional CNN or GRU layers. As shown in Table 2, the results demonstrate that CNN layers significantly improve local pattern recognition capability. (2) Addressing class imbalance and small sample issues in the dataset, we implemented multiple data augmentation approaches. Table 2 shows the Border-SMOTE strategy achieved optimal performance.

(3) For better multimodal fusion in branch models, we introduced the Q-former fusion method. The "† + Q-former" results in Table 2 validate its effectiveness. (4) To further enhance branch model performance, we developed Personality-aware Audio Map Text Feature. After obtaining different audio-map-text features through modality mapping model training, various text feature fusion methods were tested. As shown in Table 2, the "‡ + Map2 RoBERTa + Cross-Att" combination delivered the best performance. Here, "Map2 RoBERTa" denotes the Pseudo-Text Feature obtained by training the corresponding LSP model, and "Cross-Att" represents the cross-attention fusion method between Pseudo-Text Feature and Personalized Text Feature.

Table 3: Final result on the test set

Training Stage	**Final Score**	**Final F1**	**Final Acc.**
Stage One	85.39	85.43	85.35
Stage Two	94.51	94.27	94.76

Consistency-aware individual-level feature integration results. In the second stage, we introduce a hierarchical fusion mechanism to integrate task-specific features across branches and temporal samples, based on the first-stage branch models. As shown in Table 3, this two-stage framework significantly improves performance: Final F1 increases from 85.43% to 94.27% (+8.84), Accuracy from 85.35% to 94.76% (+9.41), and Final Score from 85.39% to 94.51%. These results validate the effectiveness of the proposed hierarchical fusion mechanism within the two-stage training framework.

4 Conclusion

In this work, we propose HOPE, a novel hierarchical multimodal framework for depression estimation that explicitly incorporates personality-aware representations and subject-level consistency modeling. By introducing the LSP module, HOPE effectively bridges the semantic gap between audio features and textual representations under privacy-constrained settings. The hierarchical fusion strategy integrates heterogeneous modalities and sample-level predictions into coherent subject-level outputs. HOPE achieved 1st place in the ACM MM 2025 MPDD Challenge (Young Track), demonstrating outstand performance. Our findings underscore the importance of modeling individualized behaviors in multimodal affective computing and open avenues for future work on personality-informed mental health assessment.

Acknowledgments

This work was supported in part by the Key R&D Program of Zhejiang (2025C01104), in part by the Scientific Research Staring Foundation of Hangzhou Institute for Advanced Study (2024HIASC2001), and in part by Zhejiang Provincial Natural Science Foundation of China (No. LQN25F020001).

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